MOTOR: Musculoskeletal Optimisation Through Open Robotics Simulation

J.P. Kardolus, Babon Innovations B.V., Utrecht, The Netherlands

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Abstract

Ground reaction forces, joint moments and muscle forces say more about how a body is loaded than joint angles do, but they still come from laboratory equipment or from slow musculoskeletal software that is tied to one kind of input. We describe MOTOR, an open musculoskeletal solver designed to take a moving body from any source that gives a skin mesh, a rotation matrix per segment and the joint centres, and to return the movement of a full-body musculoskeletal model (MyoFullBody, 416 muscles) and the forces that move it. Its first part fits the model to the body frame by frame: the pose closest to the input body with every joint within its anatomical limits, the segment rotations as the main target, each coupled to the group of bones that covers the same piece of body, about a hundred virtual markers on anatomical landmarks placing the joint centres, and the bones held inside the body's skin. The fit's constants are solved in three levels: the landmarks' offsets and weights once over many synthetic bodies, each person's own constants once over frames from all of that person's recordings, and the pose per frame. Its second part solves the whole trial as one optimisation of the motion, the ground contact and the pelvis residual on a single GPU with Newton and MuJoCo Warp; adding the muscles to that solve is still being built. This working draft gives the method and the validation protocol; MOTOR's own results are not in yet. The protocol runs MOTOR inside a monocular pipeline (one smartphone video, the body estimator GEM-X, then MOTOR) on the OpenCap LabValidation cohort against marker-based motion capture, force plates and EMG, with subjects 7 to 11 held out from every choice of setting. What is measured so far is the per-frame chain that now gives the whole-trial solve its starting point: on OpenCap Monocular's matched envelope its joint angle error is 5.16°, its vertical ground reaction force error 16.2% of body weight, and its pelvis residual stays within the limits on only 21% of frames.

Gap: MOTOR's headline numbers go here: its pooled joint angle error on the matched envelope, its vertical and anterior ground reaction force error, its joint moment error against OpenCap's, and its time per trial on one GPU against OpenCap's recorded solves, all from the whole-trial solver on the 720 single-camera inputs with the held-out subjects 7 to 11 apart. They wait on the solver's muscles and batching (#98) and its full run.

MOTOR is open source under the MIT licence (Section 5).

1 Introduction

What a movement does to the body is carried by forces. The ground reaction force under each foot, the net moment at each joint and the force in each muscle describe how a knee is loaded in a squat, how hard a calf works in push-off, or how one leg spares the other after an injury; joint angles describe only the shape of the movement that produced them. These forces are measured in a gait laboratory with force plates and marker-based motion capture, and the muscle forces, which no instrument measures directly, are estimated from that data by a musculoskeletal model in software

such as OpenSim [1–3] or its optimal-control extension Moco [4]. The laboratory, its operators and its solve times keep these quantities out of most clinics, sports settings and studies [5].

OpenCap moved this analysis to smartphones [6]. From two or more synchronised phone videos it estimates keypoints, fits a scaled OpenSim model by inverse kinematics on virtual markers, and solves a muscle-driven tracking simulation for the ground reaction forces, joint moments and muscle activations. On its LabValidation cohort it recovers joint angles within a few degrees of motion capture, and kinetics close enough to tell clinically relevant patterns apart, such as an asymmetric squat. OpenCap Monocular brings the analysis to a single video [7]. OpenCap's own recorded dynamic solves in its LabValidation release take 44 to 70 minutes per trial, and the pipeline is built around its own keypoints and model.

Others have moved musculoskeletal fitting onto the GPU and toward a body model. Cotton fitted biomechanical models to video as differentiable models in MuJoCo [8, 9]; the Portable Biomechanics Laboratory fits one body scale jointly across all of a person's trials from a handheld phone [10]; KinTwin recovers joint torques, muscle activations and ground reaction forces from markerless capture [11]; and the Biomechanical 3D Body fits a person's constants over frames sampled from all of their recordings and weighs its landmarks by their own residuals [12]. MOTOR takes the first of those two ideas for its per-person level; the second was tried during its development and is not its default. SKEL re-rigs a parametric body model with a biomechanical skeleton that sits inside its skin [13], HSMR estimates that skeleton from a single image [14], BioPose regresses biomechanically accurate poses from monocular video [15], and AddBiomechanics automates model scaling, inverse kinematics and inverse dynamics from motion capture [16]. On the simulation side, MyoSuite provides MuJoCo musculoskeletal bodies [17], MuJoCo Warp runs MuJoCo on the GPU [18], and Newton builds a GPU physics engine on Warp around it [19, 20]; MuscleMimic [21] and MS-Human-700 [22] train controllers for bodies of 400 to 700 muscles there, and static optimisation on MuJoCo models has run on the CPU [23, 24].

Meanwhile, parametric body models have become the common output of human motion work. Video estimators such as GEM-X [25, 26] and SAM 3D Body [27] recover a body model's mesh and per-segment rotations from video, motion generators such as Kimodo [28] produce the same rotations from text, and the SOMA convention [29] unifies several body models under one skeleton. Such a body already carries most of what a musculoskeletal fit needs: a rotation per segment, joint centres and a skin. Most musculoskeletal pipelines use little of it. Inverse kinematics on virtual markers keeps only points, so each segment's rotation about its own long axis must be recovered from markers a few centimetres apart, and nothing keeps the fitted bones inside the body they came from.

This paper describes a solver for that step: from a moving body, given as a skin mesh, segment rotations and joint centres, to the dynamics of a full-body musculoskeletal model. Its design combines four elements. First, a per-frame fit that takes the segment rotations as its main target, each coupled to the group of bones in MyoFullBody that covers the same piece of body (one spine segment against several vertebrae, for example), with about a hundred virtual markers on anatomical landmarks beside them to place the joint centres, the bones held inside the input's skin, and the joint limits and couplings held exactly. Second, a three-level solution of that fit: once over many synthetic bodies for each landmark's offset and weight, under a rule checked on bodies kept out of that tuning; once per person for the person's constants from all of their frames together; and once per frame for the pose alone. Third, the whole trial solved as one optimisation of motion, contact and residual on a single GPU with Newton and MuJoCo Warp, with the per-frame fit as its starting point; the muscles are being added to it. Fourth, a judgement of the fit on how well the model follows the body (joint positions, segment rotations, bones inside the skin), never on the marker distances the solver itself pulls on. The first, second and fourth elements are built and measured

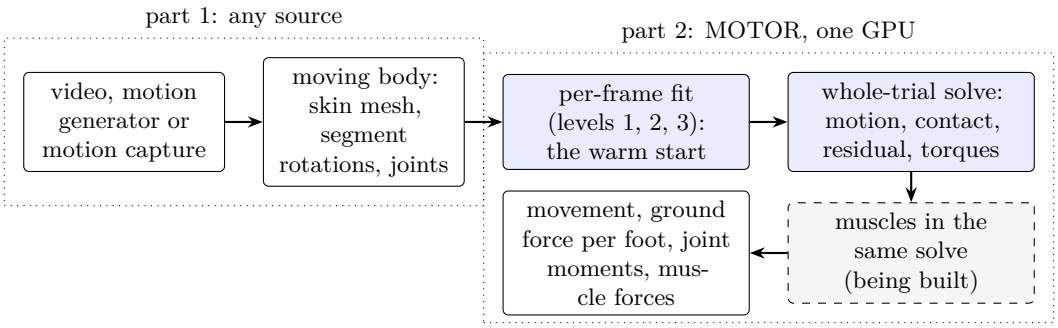


Figure 1: The pipeline. Part 1 supplies a moving body; in the validation it is GEM-X on one smartphone video. Part 2, MOTOR, fits MyoFullBody to that body frame by frame and then solves the whole trial at once on one GPU. The dashed step, muscles inside the whole-trial solve, is being built (Section 2.6).

inside the per-frame chain; the third has run only during its development, and MOTOR's validation as a whole has not run. To our knowledge the per-frame chain is the first open chain from video to muscle forces (fit, residual reduction, static optimisation) on Newton and MuJoCo Warp.

The validation protocol (Section 2.8) runs MOTOR inside a monocular pipeline, the way most of its users would meet it: one smartphone video, GEM-X, then MOTOR, on the OpenCap LabValidation cohort [6], against marker-based inverse kinematics, force plates and EMG, and compared with OpenCap on its own cohort and reference. This draft reports the method, the protocol and the per-frame chain's full validation; MOTOR's own results are gaps until its full run. The same solver runs inside SCORE, an open environment synthesis framework, on Kimodo's generated motion; other estimators and motion capture fit the same input, and their validation is left to future work.

2 Methods

2.1 Pipeline overview

The pipeline has two parts (Figure 1). The first turns sensor data, or a generator's output, into a moving body: per frame a skin mesh, a rotation matrix per segment of a kinematic chain, and that chain's joint centres. The second, MOTOR, turns that body into the movement of a musculoskeletal model and the forces that move it. This paper is about the second part. The first only supplies inputs; any source that gives the three parts can serve in that role, though only GEM-X has been run.

In the validation, the first part is GEM-X [25, 26] on one smartphone video. GEM-X gives a SOMA body [29] per frame: its identity (shape), root translation and global segment rotations, at 30 Hz, the rate to which GEM-X resamples every clip. MOTOR takes this body as given and never changes its shape or pose. GEM-X is given the camera's field of view, which MoGe-2 [30] estimates from the same video, and its own contact post-process holds the standing feet still. GEM-X's output has no floor, so the floor is estimated from the video: a plane is fitted to MoGe-2's points of the floor beside the person, the whole clip is rotated rigidly so that this plane is level, and the clip is lowered until the lowest point of its feet touches it.

2.2 The musculoskeletal model

The model is MyoFullBody from MyoSuite [17], its arm and hand from MoBL-ARMS [31, 32]: 416 muscles, 86 bones and 132 joints, of which one is the free root, 36 are coupled to another joint by a polynomial (the knee's and the shoulder girdle's, for example) and 95 are free to move on their own. Its bone groups (below) are sized to the person's SOMA body, each group's length taken from

its SOMA segment, joint to joint; MOTOR does not solve for any length. Its masses are SOMA's segment volumes spread over the subject's measured mass.

2.3 Segments, bone groups and landmarks

Segment rotations and bone groups. Each SOMA segment is coupled to a *bone group*: the bones of MyoFullBody that cover the same piece of body, one of which, the group's *frame bone*, carries the segment's rotation (Appendix D). A spine segment is coupled to the vertebrae of its height, paired by their share of the spine's height; the hand to the bones of the wrist and hand, the lunate carrying its rotation. Each segment drives one group. SOMA has two neck segments where MyoFullBody has one rigid neck: the first drives it, and the second drives nothing, though its rotation error is still reported.

How a segment's frame lines up with its group's frame is read once from the two models at rest (SOMA's template in its bind pose, MyoFullBody in its default pose with its couplings applied) and kept as a fixed *rest alignment* per group. Each group's anatomical axes are built on both models from the same anatomy: the line between the group's end joints or between landmarks, the bone's own axis, or a hinge line, such as the line through the first and fifth metatarsal heads that the foot and toes share. The feet are then turned about that hinge line until SOMA's foot, flat on its floor at rest, is MyoFullBody's heel bone flat in its default pose, and the groups whose bones would stand outside SOMA's skin along those lines (the head first) are turned and shifted rigidly at rest to their least depth outside it. The rest alignment is geometry of the two models: it is never fitted to motion data or to any reference joint angle, and never per person or per clip.

Virtual markers on anatomical landmarks. Beside the rotations the fit takes virtual markers on anatomical landmarks, each defined by its anatomy following the ISB recommendations [33, 34]: the acromion, the humeral epicondyles, the anterior and posterior superior iliac spines, the malleoli, the metatarsal heads and so on. Each landmark sits on a fixed point of the SOMA mesh, given as a face and barycentric weights, so it moves with the person's skin, and on the matching point of a MyoFullBody bone, plus an offset from the bone toward the skin. A study of the two models placed 102 landmarks; the fit uses 84 of them, each with its own weight. No licensed dataset's marker list is used.

Joint centres as segment ends. SOMA's rig joints mark where each bone group starts and ends: the hips, knees, ankles, shoulders, elbows, wrists and the joints between spine segments. Each such end is one more marker, at MyoFullBody's own joint centre there, aimed at SOMA's joint. The joints inside a group (the lumbar levels, the shoulder blade, the wrist bones, the joints of the inner foot) are not aimed at anything; the landmarks and the model's couplings place them. No joint gets a special case.

The skin. The bones belong inside the body they came from. SKEL builds this idea into a body model, a biomechanical skeleton placed inside the skin of SMPL [13]; MOTOR uses the input's own skin as a target instead. Each frame's SOMA skin is posed, and a fixed subset of the vertices of every bone mesh is measured against it: each vertex's offset to the closest point of the skin, or zero when the vertex is inside, as decided by the generalised winding number [35].

2.4 The per-frame fit

For one frame, with the person's constants fixed, the fit finds MyoFullBody's coordinates q that come closest to the input body while every joint stays within its anatomical limits:

$$\min_{q} \frac{1}{N_g}\sum_{g}\left\|\log\!\left(\hat{R}_g^{\top} R_g(q)\right)\right\|^2 \;+\; \frac{\lambda_m}{\sum_m w_m}\sum_{m} w_m\left\|p_m(q)-\hat{x}_m\right\|^2 \;+\; \frac{\lambda_s}{N_b}\sum_{b}\left\|d_b(q)\right\|^2 \qquad (1)$$
$$\text{s.t. } q_{\text{follower}} = f(q_{\text{leader}}) \text{ for every coupling}, \qquad q_{\min} \le q_{\text{leader}} \le q_{\max}.$$

The first term is the main one: for each of the N_g bone groups, the angle between the frame bone's orientation $R_g(q)$ and its SOMA segment's rotation $\hat{R}_g$ composed with the group's rest alignment. The second pulls each marker m (the landmarks and the segment ends) at its point $p_m(q)$ on the model toward its point $\hat{x}_m$ on the SOMA body, weighed by its weight w_m. The third holds the N_b bone vertices inside the skin, d_b being a vertex's offset outside it. Each goal is normalised by its own count, as OpenSim's inverse kinematics weighs its goals [2]. The rotations are the main term; the markers' goal λ_m is set by level 1 (2.1169), and the skin goal λ_s is 10^3, chosen in a sweep that traded the bones outside the skin against the rotation error. A coupled joint (a *follower*) moves with a *leading* coordinate by MyoFullBody's own polynomial f, held as an equality constraint; the limits are the published ranges of the leading coordinates, held as bounds. The fingers are held at SOMA's rest.

The reference solver is mink [36], a differential inverse-kinematics library on MuJoCo. Each of its steps solves one quadratic programme: the segment rotations and the marker positions are tasks on frames of the model, the couplings are equality constraints, the leading coordinates' ranges are limits, and the finger joints are frozen; the step length is controlled as in Levenberg–Marquardt. The skin term adds three rows per bone vertex, its offset from the closest point of the skin, whose Jacobian keeps only the motion away from the closest face, edge or vertex of the skin mesh. On the GPU the same step is a damped Gauss–Newton step in the leading coordinates, solved for every frame at once with Warp's tile Cholesky factorisation [20]. On every frame of a development clip it matches mink within 0.0025° in every joint angle and 0.0042 mm in every model point. A frame that does not converge in 300 steps stops the run with an error; it is never kept silently.

2.5 Solving in three levels

The fit has constants that do not change from frame to frame: where each landmark sits and how much it is weighted, and where the joints inside each bone group sit in this person. They are solved in three levels, each held fixed for the levels that follow, and every level is judged on how well the model follows the body, never on the marker distances the solver pulls on.

Level 1, once over many bodies. One bounded least-squares problem tunes, all together: each landmark's offset from its bone (for a landmark that defines a joint, only along its bone); each landmark's weight, left and right mirrored and shared; each segment end's position on MyoFullBody, within 40 mm of the model's own joint centre; and two global weights, the markers' goal λ_m and the segment ends' weight among the markers. Its objective has two terms, each normalised by its value at the start so that the two weigh alike: each body region's bones' depth outside the SOMA skin, and each segment's rotation error against its bone group. Each evaluation runs levels 2 and 3 on every tuning body. The bodies are synthetic SOMA bodies (tall, short, heavy, slim, long- and short-legged) doing seven clips each (a Kimodo walk, a rig walk, arm raises, a reach, a squat, running and sit-to-stand). They are split by body into tuning and check bodies, every clip of a body on one side. The check bodies are scored during the run and never used to choose, and a

tuned point is accepted only when they score within 10% of the tuning bodies. Level 1's gradient is taken through levels 2 and 3 by implicit differentiation; the columns that path does not yet cover are taken by finite differences, and every column is checked against finite differences before a run is used. Level 1 sees only synthetic bodies' skin depth and rotation error, never any reference joint angle.

Level 2, once per person. With level 1 fixed, the person's constants are solved from all of their frames together [12]. About 30 frames, spread over the person's clips by their length, are posed, and one sparse least-squares problem solves, together with those frames' coordinates, each landmark's point on its bone and the position of each joint inside a bone group, each within 3 cm of where level 1 and the model put it and drawn back toward it by a weight. Only what the data can tell apart is solved: a joint position whose effect on the residuals is a combination of the other unknowns' effects stays where the model has it. The landmarks on the soles' pads and on the skull never move, so a foot cannot sink through the sole of the skin nor the skull through the head. Level 2 has no skin term: with one, the solve moved the inside joints by 40 to 52 mm to lower the bones' depth outside the skin instead of fitting the person. The constants are written into the person's model file, so every later solve uses them.

Level 3, per frame. With the person's constants fixed, every frame solves only its pose, by Equation (1): the rotations as the main term, the markers beside them, the bones inside the skin, and the joint limits and couplings as hard constraints.

2.6 The whole-trial solve

The per-frame fit follows the body but knows nothing of forces: it can put a foot where no ground could hold it, and the frames it gives need not be consistent with the forces that would move the model between them. MOTOR therefore solves the whole trial as one optimisation, with the per-frame fit as its starting point (its *warm start*). The movement, the dynamics with contact and residual, and the contact's physical settings are built and described here; the muscles, batching across trials and its use inside SCORE are being built.

The movement. Every leading coordinate's path over the trial is one uniform cubic B-spline with one control point per frame, so speeds and accelerations are the spline's own derivatives and there are no kinematic defects to constrain. The root's position is a spline like a coordinate's. Its orientation is the per-frame fit's orientation, low-pass filtered, times a correction rotation whose rotation vector, in the body frame, is again a spline. The movement's cost is the fit's own cost of Equation (1) over every frame at once, plus the spline's jerk, with the leading coordinates inside their ranges by bounds on their control points. The jerk weight of each coordinate is not tuned: it is the one that puts the solve's smoothing at the 6 Hz cut-off the per-frame chain filters with [5]. Each frame's Gauss–Newton block comes from the GPU fit's own pieces, and the normal equations are banded, since each frame's rows touch only its three nearest control points.

Dynamics, contact and the residual. The *dynamics solve* adds the dynamics of every frame to the same unknowns. MuJoCo Warp, inside Newton, gives the generalised forces $\tau_{\text{needed}}(q, \dot{q}, \ddot{q})$ that each frame's motion needs. The ground's force comes from the motion alone, with no force unknowns: every point of the person's skin within 2 cm of its floor is pushed by a smooth spring and damper of the Hunt–Crossley kind with a smoothed Coulomb friction, the form of OpenCap's

tracking simulation [6],

$$f_n = k\,\sigma_s(-h)\,\sigma\big(1 - c\,v_n\big), \qquad \mathbf{f}_t = -\mu\,f_n\,\frac{\mathbf{v}_t}{\sqrt{\|\mathbf{v}_t\|^2 + \varepsilon^2}}, \tag{2}$$

with h the point's height over the floor, v_n and $\mathbf{v}_t$ its normal and tangential speed, k the stiffness, c the dissipation, μ the friction, and σ_s and σ smoothed ramps, so that a lifting foot pushes nothing and never pulls. The forces left for the body to supply, on the leading coordinates, are

$$\tau = G^\top\big(\tau_{\text{needed}} - J_c^\top f_c\big), \tag{3}$$

where G maps the leading coordinates' velocities to all coordinates' (the couplings' Jacobian), f_c stacks the contact points' forces and J_c is the Jacobian of their positions. The root's six rows of τ are the pelvis residual, the force and moment that neither the body nor the ground explains. An augmented Lagrangian holds them on every frame under 2% of body weight in force and 1% of body weight times the centre of mass height in moment; the moment limit is that of [37], and the force limit is our own, tightened from 5% with AddBiomechanics' results [16] as the benchmark. The other rows are the joints' torques, kept small as an effort term.

The whole is solved by Levenberg–Marquardt on the banded normal equations. The Jacobians of the needed forces come from three places: MuJoCo's mass matrix for the accelerations (exact), the joints' damping for the speeds, and one-sided finite differences on the GPU for the positions. Before the first iteration the contact is settled: where a frame's lowest skin point lies under the floor at the warm start, that frame's floor is lowered to it. Without this, a seat whose height comes from the protocol, while the video places the person's skin lower, pushed the body with many times its weight.

Settings. The contact's settings are physical: a whole foot's stiffness of 6×10^6 N/m spread over its skin points (OpenCap's six contact spheres per foot at 10^6 N/m each), a dissipation of 2.0 s/m and a friction of 0.8 as in OpenCap's contact spheres, and a slip transition ε of 0.05 m/s. The effort term's weight is set so that, at the warm start, it is 0.1 of the fit's cost. These settings were tuned on subjects 2 to 5 only, against the force plates with the dynamics solve's own contact, and the tuning kept these first values (Section 3.4). Nothing is ever fitted to the reference joint angles.

Muscles. The muscles are to enter the same solve: per person, polynomial muscle–tendon lengths, whose derivatives are the moment arms, fitted to MuJoCo's own tendon lengths, and MuJoCo's muscle force curves, so that the activations are solved together with the motion and the contact rather than distributed afterwards.

> **Gap:** The muscles' formulation as built (the muscle terms, the activation dynamics and how they enter the solve) replaces this box once it is built and measured (#98).

2.7 The warm start: the per-frame chain

Before the whole-trial solve existed, MOTOR's forces came from a chain of per-frame steps, which now gives the whole solve its starting point. After the fit, the standing feet are put on the floor by moving the whole body up or down, never by changing the pose. A residual reduction in the manner of OpenSim's RRA then adjusts the pelvis, the trunk and the masses within a per-frame cap, so that the ground can explain the movement. Last, the forces are split in the two stages of OpenSim's static optimisation [1]: the ground reaction force from the pelvis' six balance equations

alone, then the muscle activations with that force fixed. Every step runs every frame at once on the GPU in Warp's deterministic mode, and two runs agree bit for bit. On LabValidation the chain's matched joint angle error is 5.16° (95% CI 4.77 to 5.59°), its vertical ground reaction force error 16.2% of body weight and its joint moment error 2.41% of weight times height. It takes a median of 66 s per trial on one NVIDIA L4 GPU, and its pelvis residual stays within both limits on only 21% of frames. Appendix B gives its full validation. These are the warm start's numbers, not MOTOR's result.

2.8 Validation protocol

The protocol below is set; MOTOR has not yet been run through it. We validate on OpenCap LabValidation [6]: nine healthy subjects (2, 3, 4, 5, 7, 8, 9, 10 and 11; subject 6 is not part of the published cohort) in walking, walking with trunk sway, squats, sit-to-stand and drop jumps, recorded with marker-based motion capture, force plates and EMG of 16 lower-limb channels. MOTOR sees only smartphone video, one camera at a time. Its reference standard is the lab's: joint angles from the markers through OpenSim inverse kinematics on the lab's model, ground reaction forces from the plates, joint moments from OpenSim's inverse dynamics remade on the lab's own model and settings, and EMG envelopes for the muscle activations.

Camera input. The video comes from five synchronised iPhones, Cam0 to Cam4 in LabValidation's numbering. The 144 non-static trials times five cameras give 720 single-camera inputs, each run on its own. For the matched comparison we use the 45° camera (Cam1) on the non-jumping activities and the fifteen joint angles that have a counterpart in OpenCap Monocular's set (Appendix C), the envelope on which OpenCap Monocular reports its results [7] and on which the author's earlier work reported GEM-X's angles [38]. The full technical validation pools all 720 inputs, every activity and every camera.

Metrics. Video and reference are not synchronised at the millisecond level, so each trial gets one time shift of the reference before scoring. Joint angles are scored by mean absolute error (MAE), centred MAE (cMAE, each trial's constant offset removed, which separates tracking from offset), root mean square error and Pearson's r, with a 95% confidence interval from a per-subject bootstrap of 20,000 draws. Ground reaction forces are scored as OpenCap scores them, in % of body weight on three axes with both feet summed, and per foot by the correlation of the vertical force. Joint moments are scored in % of weight times height over OpenCap's fifteen, with the knee adduction moment's early-stance peak apart. In squats, the vasti symmetry index is scored by its AUC against the EMG's own asymmetry label. Activations are scored by on–off agreement and correlation with each EMG channel, and the pelvis residual by the share of frames within both limits.

Held-out subjects. The only settings chosen on LabValidation are chosen on subjects 2 to 5, against their force plates and never against joint angles; subjects 7 to 11 are never used to choose anything and are reported apart. The tool that tunes MOTOR refuses held-out subjects. The protocol's record also asks that anything tuned on LabValidation be scored leaving one subject out, as the author's earlier work did [38]; the settings so far were chosen on the fixed split above, and MOTOR's full run adds the leave-one-subject-out scores.

Table 1: The headline comparison. Joint angles on the matched envelope (Cam1, no drop jumps, 15 coordinates); kinetics over all cameras and activities. GRF in % body weight, both feet summed; moments in % weight × height. OpenCap: two cameras, its published figures (OpenCap Monocular's angle MAE: 4.8°). Warm start: the per-frame chain (Section 2.7), not MOTOR's result. Held out: subjects 7–11.

	OpenCap	Warm start (chain)	MOTOR	MOTOR, held out
Joint angle MAE, matched 15 (°)	4.5	5.16	*gap*	*gap*
Joint angle cMAE, matched 15 (°)	–	3.06	*gap*	*gap*
GRF vertical MAE	11.4	16.2	*gap*	*gap*
GRF anterior MAE	3.5	7.4	*gap*	*gap*
GRF medio-lateral MAE	3.8	3.0	*gap*	*gap*
Joint moments, OpenCap's 15, MAE	1.2	2.41	*gap*	*gap*
Early-stance knee adduction peak, left	0.30	1.51	*gap*	*gap*
Vasti symmetry AUC against the EMG label	0.83	0.66	*gap*	*gap*
Frames within both residual limits	–	21%	*gap*	*gap*

Gap: Figure: MOTOR's vertical and anterior ground reaction force per foot against the force plates over the stance phase of the median walking and the median squat trial, with the warm start's beside it, and MOTOR's pooled matched MAE and cMAE with the per-subject bootstrap's 95% CI beside OpenCap Monocular's 4.8° and OpenCap's 4.5°. Waits on the whole-trial solver's full run (#98).

Figure 2: MOTOR against the force plates and the lab's joint angles (gap).

3 Results

MOTOR's validation has not run. This section gives the protocol's headline tables with MOTOR's columns as gaps, which its full run on the 720 inputs fills (#98); the full technical validation is laid out in Appendix A. What is measured so far is the warm start (Section 2.7, Appendix B) and the dynamics solve during its build (Section 3.4).

3.1 Matched comparison with OpenCap

OpenCap validates on the same cohort and reference standard, which makes it the most direct external comparison: from two videos it reports a joint angle MAE of 4.5° [6], and OpenCap Monocular reports 4.8° from the single 45° view on the non-jumping activities [7]. Table 1 sets MOTOR beside them, with the warm start as a baseline.

3.2 Accuracy regimes

Gap: Which joint angles, force axes, moments and muscles MOTOR tracks well, moderately and poorly (by held-out correlation and cMAE, from Appendix A), and how they compare with the warm start's regimes, where the elbows and toes carry constant offsets from the input and the pelvis residual stays above its limits. Waits on the full run (#98).

3.3 Speed

OpenCap's own recorded dynamic solves in its LabValidation release take 2635 s for a squat, 4096 s for a drop jump and 4196 s for a walk, on windows of 0.99 to 1.68 s with an 80-muscle model; the

Table 2: The dynamics solve during its build, on the 8 training trials (subjects 2–5): the warm start (the chain), the dynamics solve before the contact was settled, and after, with its settings tuned. "Own contact": the ground force the dynamics solve's contact model gives. "Chain's estimate": the chain's two-stage force estimate run on the dynamics solve's path. GRF in % body weight, moments in % weight × height. Development measurements, not results.

	Chain	Before settling	Settled, tuned
Trials with the residual within limits on every frame	–	5 of 8	7 of 8
Joint angle MAE (°)	7.85	9.58	8.18
GRF vertical MAE, own contact	–	28.3	21.1
GRF anterior MAE, own contact	–	13.3	9.6
GRF vertical MAE, chain's estimate	16.7	21.7	15.9
GRF anterior MAE, chain's estimate	5.3	6.1	4.9
Joint moments MAE	1.50	2.16	1.81

warm start takes a median of 66 s per trial on one NVIDIA L4 (Table 19).

Gap: MOTOR's wall time per trial on one GPU, the whole-trial solve included, per activity, beside OpenCap's recorded solves, and its bit-for-bit repeatability across two runs. Waits on batching across trials and the full run (#98).

3.4 Measured so far: the dynamics solve during its build

The dynamics solve has run only on eight training trials of subjects 2 to 5, one GPU each, while it was being built; we report it as a development measurement, not as a result (Table 2). Once the contact is settled at the warm start (Section 2.6), it holds the pelvis residual within both limits on every frame of 7 of the 8 trials, and on 98% of the sit-to-stand's frames; before the settling it did so on 5. On the protocol's measures it is not yet better than its own warm start. The joint angle error rises from the chain's 7.85° to 8.18°, and the joint moment error from 1.50 to 1.81% of weight times height. When the chain's own force estimate is run on the dynamics solve's path, the vertical force error is close to the chain's (15.9 against 16.7% of body weight). A walk takes 45 to 90 s and a squat or sit-to-stand 6 to 12 min.

Two attempts to improve the forces did not hold and were not kept. The first let a foot that the video shows in stance push over its whole sole rather than only at its lowest points (36 training trials of Cam1). A wider sole let the solve move the centre of pressure to wherever the movement's demand for a moment wanted it: as the sole's contact depth grew from 5 to 40 mm, the anterior force error rose from 7.30 to 8.09% of body weight and the centred centre-of-pressure error from 51.6 to 62.7 mm. The second used the settings that scored best on the plates (a stiffness of 2×10^7 N/m, a slip transition of 0.2 m/s, a friction of 0.6 and an effort share of 0.01). Its anterior force error fell from 9.6 to 6.2% of body weight and its joint moment error from 1.81 to 1.66%, but the sit-to-stand's residual stayed within the limits on only 1% of its frames. The landmark weights learned by the EM step of [12] lost the sit-to-stand in the same way. Both point to the same two causes: the moment the video's kinematics demand at the pelvis, and the seat's contact.

4 Discussion

4.1 A solver from any body

MOTOR takes a moving body, not a video. Its input is what parametric body models now produce: a skin mesh, a rotation per segment and the joint centres. This is meant to let one solver serve

several sources. A video estimator such as GEM-X supplies it from one phone, as in the validation; a motion generator such as Kimodo supplies it from text, and SCORE runs MOTOR on Kimodo's motion to give its generated people forces and muscles; motion capture can supply it through a body model fitted to the markers; and the SOMA convention [29], which recovers one skeleton from the posed meshes of several body models, opens the route from other estimators. Only GEM-X has been run on LabValidation. A second estimator through the same solver, such as SAM 3D Body [27], as the author's earlier work did for joint angles [38], is the test that this claim still needs.

The choice of inputs is also what makes the fit well posed. A segment rotation fixes a bone group's rotation about its own long axis, which markers a few centimetres apart resolve poorly, so the rotations are the main term and the markers place the joint centres beside them. The skin gives the fit a target that a marker set does not: the bones must lie inside the body they came from, a constraint SKEL builds into its body model [13] and MOTOR takes from the input. And the fit is judged on what it is for, how well the model follows the body, rather than on the marker distances it pulls on, which a solver can always lower by trading one landmark against another.

Running every frame and every step on one GPU changes what a solve costs. The warm start already solves a trial in a median of about a minute where OpenCap's recorded solves take 44 to 70 minutes; the models and hardware differ (the hardware of OpenCap's solves is not recorded), and the whole-trial solve adds its own time, 45 s to 12 min per trial during its build. Determinism matters as much: in Warp's deterministic mode two runs of the chain agree bit for bit, which answers, for this chain, the CPU against GPU drift reported on MyoSuite bodies [39].

4.2 Implications

Gap: What MOTOR's measured accuracy and speed mean for clinics, sports and studies that have no laboratory, and for training data made from generated motion, written once its full run (#98) says how close it comes to OpenCap on the same cohort.

4.3 Camera sampling and fast movements

GEM-X resamples every clip to 30 Hz before MOTOR sees it, so the fastest parts of a movement, an impact or a push-off, are sampled at that rate whatever the phone recorded. The drop jumps show it in the warm start: of the five activities they have the highest joint moment error (3.59% of weight times height, against 1.55 to 1.88 for the others), the largest pelvis residual (a median of 10.9% of body weight, its 95th percentile 85.5%), and two of the five inputs that failed did so because a drop jump frame's fit did not converge. MOTOR's dynamics solve has run on one drop jump only, among its build trials, where it diverged until the contact was settled; whether the whole-trial solve holds the landing is part of its full run.

4.4 Limitations

The solver is not yet validated. The muscles and batching across trials are still being built, and the dynamics solve has run only on eight training trials, where it does not yet improve on its warm start. Every claim about MOTOR's accuracy and speed waits on its full run.

What the input carries. MOTOR follows its input, so the input's errors pass through. In the warm start the elbows come out straighter than the lab's in every one of the 89 envelope trials, by a mean 21.1° (right) and 23.7° (left), with centred errors of 3.0 and 4.6°. On 540 walking arm trials

GEM-X's own elbow is a median 18.5° straighter than the markers', and the fit's elbow differs from GEM-X's by a median 2.5°. The pelvis tilt is offset in one direction in 82 of 89 envelope trials, by a mean 5.3°. The lengths are SOMA's, so an error in the estimated body shape becomes an error in the model's size. Monocular depth ambiguity limits the out-of-plane rotations, as it does for every single-camera method: in the recorded runs, the knee of the leg far from the camera sits 37 to 44 mm from the centre of pressure in the frontal plane where the lab has 9 to 16 mm, which accounts for the error in the early-stance knee adduction moment.

Forces and residuals. In the warm start the pelvis residual stays above its limits (medians of 7.0% of body weight and 8.9% of weight times centre of mass height), the anterior ground reaction force error is about twice OpenCap's, and the activation pattern is partly off (rectus femoris does not follow its EMG, $r = -0.08$). Holding the residual is what the whole-trial solve is for; whether it does so while keeping the forces and moments right is what its validation will show. The force limit itself, 2% of body weight, is our own and stricter than the 5% the chain started from, so the share of frames within the limits is not comparable with a study that uses another limit.

The fit's own measures. Level 1 tunes on synthetic SOMA bodies, not on people. The last level 1 run whose held-out score is recorded met its rule: its check bodies' score fell from 4.26 to 3.56 while the tuning bodies' fell from 4.53 to 3.44, and the rule's comparison, 3.56 against the tuning bodies' 3.51, was within 10%. The tuned point the fit uses now came from a later run with a broader objective, whose held-out score is not yet recorded. The bones are not yet all inside the skin: on the fifteen clips of the synthetic body of mean shape, 5.15% of the bone points lie outside it (averaged over body regions), against a target of under 5%.

One cohort. LabValidation is nine healthy adults in a laboratory. MOTOR holds every joint within the published healthy ranges; people whose joints move past or short of them, as many patients do, need ranges of their own, which MOTOR does not yet solve.

4.5 Relationship to adjacent work

MOTOR shares its aims with OpenCap [6, 7], from which it takes its cohort, its reference standard and its contact model's form; it differs in taking a body model's rotations and skin as input and in solving on the GPU. Cotton's group showed first that biomechanical models can be fitted from video in MuJoCo on the GPU [8], that one body scale can be fitted across all of a person's trials [10], that torques and forces can be recovered from markerless capture [11], and that a person's constants can be fitted over frames sampled from all of their recordings [12], which MOTOR's level 2 does. SKEL [13] and AddBiomechanics [16] put a biomechanical skeleton inside a body model and fit models to motion capture; MOTOR keeps the input body as given and fits the model inside it. Among monocular methods, HSMR [14] regresses SKEL's skeleton from one image and BioPose [15] regresses biomechanically accurate poses from video; both stop at the pose, where MOTOR goes on to forces. MuscleMimic [21], MS-Human-700 [22] and LocoMuJoCo [40] use musculoskeletal bodies on MuJoCo and MuJoCo Warp to learn controllers; MOTOR solves for the forces of a given motion. The author's earlier work reads clinical joint angles straight from the same body model's rotations [38]; MOTOR uses none of that method and goes on to forces and muscles.

5 Conclusion

MOTOR is an open musculoskeletal solver designed to take a moving body from any source, a skin mesh, segment rotations and joint centres, and drive MyoFullBody with it: one fit per frame with the segment rotations as the main target, anatomical landmarks beside them, the bones inside the skin and every joint within its limits, solved in three levels, and then the whole trial as one optimisation of motion, contact and residual on a single GPU with Newton and MuJoCo Warp. Its warm start, the per-frame chain, is validated in full on LabValidation, solves a trial in a median of about a minute and repeats bit for bit; MOTOR itself is not yet validated.

Gap: The conclusion's measured sentence goes here: MOTOR's accuracy against OpenCap on the same cohort and reference, on the held-out subjects, and its time per trial. Waits on the whole-trial solver's full run (#98).

Forces and muscle activity have been out of reach wherever there is no laboratory. A solver that takes the body any modern estimator or generator produces, and returns the forces quickly on one GPU, is meant to bring them to wherever a movement can be filmed or generated.

Data and code availability

MOTOR, its validation tools and this paper's sources are open under the MIT licence at https://github.com/Babon-Innovations-b-v/score; every number in this paper is read from a file or record named in a comment beside it in the paper's source. We collected no new data from people. LabValidation is the public release of OpenCap's authors [6], used under its Apache-2.0 licence and fetched on demand, never redistributed.

References

[1] Scott L. Delp, Frank C. Anderson, Allison S. Arnold, Peter Loan, Ayman Habib, Chand T. John, Eran Guendelman, and Darryl G. Thelen. OpenSim: Open-source software to create and analyze dynamic simulations of movement. *IEEE Transactions on Biomedical Engineering*, 54 (11):1940–1950, 2007.

[2] Ajay Seth, Jennifer L. Hicks, Thomas K. Uchida, Ayman Habib, Christopher L. Dembia, James J. Dunne, et al. OpenSim: Simulating musculoskeletal dynamics and neuromuscular control to study human and animal movement. *PLOS Computational Biology*, 14(7):e1006223, 2018.

[3] Apoorva Rajagopal, Christopher L. Dembia, Matthew S. DeMers, Denny D. Delp, Jennifer L. Hicks, and Scott L. Delp. Full-body musculoskeletal model for muscle-driven simulation of human gait. *IEEE Transactions on Biomedical Engineering*, 63(10):2068–2079, 2016. doi: 10.1109/TBME.2016.2586891.

[4] Christopher L. Dembia, Nicholas A. Bianco, Antoine Falisse, Jennifer L. Hicks, and Scott L. Delp. OpenSim Moco: Musculoskeletal optimal control. *PLOS Computational Biology*, 16(12): e1008493, 2020. doi: 10.1371/journal.pcbi.1008493.

[5] David A. Winter. *Biomechanics and Motor Control of Human Movement*. Wiley, 4 edition, 2009.

[6] Scott D. Uhlrich, Antoine Falisse, Łukasz Kidziński, Julie Muccini, Michael Ko, Akshay S. Chaudhari, Jennifer L. Hicks, and Scott L. Delp. OpenCap: Human movement dynamics from smartphone videos. *PLOS Computational Biology*, 19(10):e1011462, 2023.

[7] Selim Gilon, Emily Y. Miller, and Scott D. Uhlrich. OpenCap Monocular: 3D human kinematics and musculoskeletal dynamics from a single smartphone video. *arXiv preprint arXiv:2603.24733*, 2026.

[8] R. James Cotton. Differentiable biomechanics unlocks opportunities for markerless motion capture. *arXiv preprint arXiv:2402.17192*, 2024.

[9] Emanuel Todorov, Tom Erez, and Yuval Tassa. MuJoCo: A physics engine for model-based control. In *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, pages 5026–5033, 2012.

[10] J. D. Peiffer, Kunal Shah, Irina Djuraskovic, Shawana Anarwala, Kayan Abdou, Rujvee Patel, Prakash Jayabalan, Brenton Pennicooke, and R. James Cotton. Portable biomechanics laboratory: Clinically accessible movement analysis from a handheld smartphone. *arXiv preprint arXiv:2507.08268*, 2025.

[11] R. James Cotton. KinTwin: Imitation learning with torque and muscle driven biomechanical models enables precise replication of able-bodied and impaired movement from markerless motion capture. *arXiv preprint arXiv:2505.13436*, 2025.

[12] R. James Cotton, J. D. Peiffer, Lucinda Williamson, John Leske, and Georgios Pavlakos. Biomechanical 3D body: Self-supervised distillation of biomechanical pose from a 3D body foundation model. *arXiv preprint arXiv:2608.29928*, 2026.

[13] Marilyn Keller, Keenon Werling, Soyong Shin, Scott Delp, Sergi Pujades, C. Karen Liu, and Michael J. Black. From skin to skeleton: Towards biomechanically accurate 3D digital humans. *ACM Transactions on Graphics*, 42(6), 2023.

[14] Yan Xia, Xiaowei Zhou, Etienne Vouga, Qixing Huang, and Georgios Pavlakos. Reconstructing humans with a biomechanically accurate skeleton. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 5355–5365, 2025.

[15] Farnoosh Koleini, Muhammad Usama Saleem, Pu Wang, Hongfei Xue, Ahmed Helmy, and Abbey Fenwick. BioPose: Biomechanically-accurate 3D pose estimation from monocular videos. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 6330–6339, 2025.

[16] Keenon Werling, Nicholas A. Bianco, Michael Raitor, Jon Stingel, Jennifer L. Hicks, Steven H. Collins, Scott L. Delp, and C. Karen Liu. AddBiomechanics: Automating model scaling, inverse kinematics, and inverse dynamics from human motion data through sequential optimization. *PLOS ONE*, 18(11):e0295152, 2023.

[17] Vittorio Caggiano, Huawei Wang, Guillaume Durandau, Massimo Sartori, and Vikash Kumar. MyoSuite: A contact-rich simulation suite for musculoskeletal motor control. *arXiv preprint arXiv:2205.13600*, 2022.

[18] Google DeepMind and NVIDIA. MuJoCo Warp: a GPU-optimized version of the MuJoCo physics simulator, 2026. URL https://github.com/google-deepmind/mujoco_warp. Apache-2.0.

[19] The Newton Developers. Newton: a GPU-accelerated physics simulation engine, 2026. URL https://github.com/newton-physics/newton. Version 1.7.0 used here, with MuJoCo Warp 3.15.0 and Warp 1.18.0. Apache-2.0.

[20] Miles Macklin. Warp: A high-performance Python framework for GPU simulation and graphics, 2022. URL https://github.com/NVIDIA/warp. Version 1.18.0 used here. Apache-2.0 since 1.6.2.

[21] Chengkun Li, Cheryl Wang, Bianca Ziliotto, Merkourios Simos, Jozsef Kovecses, Guillaume Durandau, and Alexander Mathis. Towards embodied AI with MuscleMimic: Unlocking full-body musculoskeletal motor learning at scale. *arXiv preprint arXiv:2603.25544*, 2026.

[22] Yunyue Wei, Chenhui Zuo, Shanning Zhuang, Haixin Gong, Yaming Liu, and Yanan Sui. Scaling whole-body human musculoskeletal behavior emulation for specificity and diversity. *arXiv preprint arXiv:2603.29332*, 2026.

[23] Jenishkumar Chauhan, Samarth Brahmbhatt, and Vineet Vashista. Towards real-world identification of fatigued muscle groups via musculoskeletal simulation. *arXiv preprint arXiv:2605.26151*, 2026.

[24] Sagar Verma. emg2tendon: From sEMG signals to tendon control in musculoskeletal hands. *arXiv preprint arXiv:2508.08269*, 2025.

[25] NVIDIA. GEM-X: Monocular whole-body 3D human pose estimation using the SOMA body model, 2026. URL https://github.com/NVlabs/GEM-X. Apache-2.0.

[26] Jiefeng Li, Jinkun Cao, Haotian Zhang, Davis Rempe, Jan Kautz, Umar Iqbal, and Ye Yuan. GENMO: A GENeralist model for human MOtion. *arXiv preprint arXiv:2505.01425*, 2025.

[27] Xitong Yang, Devansh Kukreja, Don Pinkus, others, and Kris Kitani. SAM 3D Body: Robust full-body human mesh recovery. *arXiv preprint arXiv:2602.15989*, 2026.

[28] Davis Rempe, Mathis Petrovich, Ye Yuan, Haotian Zhang, Xue Bin Peng, et al. Kimodo: Scaling controllable human motion generation. *arXiv preprint arXiv:2603.15546*, 2026.

[29] Jun Saito, Jiefeng Li, Michael de Ruyter, Miguel Guerrero, et al. SOMA: Unifying parametric human body models. *arXiv preprint arXiv:2603.16858*, 2026.

[30] Ruicheng Wang et al. MoGe-2: Accurate monocular geometry with metric scale and sharp details. *arXiv preprint arXiv:2507.02546*, 2025.

[31] Katherine R. S. Holzbaur, Wendy M. Murray, and Scott L. Delp. A model of the upper extremity for simulating musculoskeletal surgery and analyzing neuromuscular control. *Annals of Biomedical Engineering*, 33(6):829–840, 2005. doi: 10.1007/s10439-005-3320-7.

[32] Katherine R. Saul, Xiao Hu, Craig M. Goehler, Meghan E. Vidt, Melissa Daly, Anca Velisar, and Wendy M. Murray. Benchmarking of dynamic simulation predictions in two software platforms using an upper limb musculoskeletal model. *Computer Methods in Biomechanics and Biomedical Engineering*, 18(13):1445–1458, 2015. doi: 10.1080/10255842.2014.916698.

[33] Ge Wu, Sorin Siegler, Paul Allard, et al. ISB recommendation on definitions of joint coordinate system of various joints for the reporting of human joint motion — part I: ankle, hip, and spine. *Journal of Biomechanics*, 35(4):543–548, 2002.

[34] Ge Wu, Frans C. T. van der Helm, H. E. J. (DirkJan) Veeger, et al. ISB recommendation on definitions of joint coordinate systems of various joints for the reporting of human joint motion — part II: shoulder, elbow, wrist and hand. *Journal of Biomechanics*, 38(5):981–992, 2005.

[35] Alec Jacobson, Ladislav Kavan, and Olga Sorkine-Hornung. Robust inside-outside segmentation using generalized winding numbers. *ACM Transactions on Graphics*, 32(4):33:1–33:12, 2013.

[36] Kevin Zakka. mink: Python inverse kinematics based on MuJoCo, 2024. URL https://github.com/kevinzakka/mink. Apache-2.0.

[37] Jennifer L. Hicks, Thomas K. Uchida, Ajay Seth, Apoorva Rajagopal, and Scott L. Delp. Is my model good enough? best practices for verification and validation of musculoskeletal models and simulations of movement. *Journal of Biomechanical Engineering*, 137(2):020905, 2015. doi: 10.1115/1.4029304.

[38] Joey Páolo Kardolus, Daan Hendriks, and Jaap Jansen. Direct clinical joint angle extraction from parametric body model rotation matrices. *arXiv preprint arXiv:2607.17639*, 2026.

[39] M. Colley. Direct MyoSuite CPU/CUDA muscle rollouts drift at the wrist with contacts disabled. MuJoCo Warp issue 1835, 2026. URL https://github.com/google-deepmind/mujoco_warp/issues/1835.

[40] Firas Al-Hafez, Guoping Zhao, Jan Peters, and Davide Tateo. LocoMuJoCo: A comprehensive imitation learning benchmark for locomotion. *arXiv preprint arXiv:2311.02496*, 2023.

A Full technical validation of MOTOR

Laid out as Appendix B of [38]: all 720 single-camera inputs (every activity including drop jumps, each camera its own input), every joint angle MOTOR outputs, ground reaction forces on all three axes per activity, joint moments, residuals, every EMG channel, per activity and per camera, with the held-out subjects 7–11 apart. Every table waits on the whole-trial solver's full run (#98); Appendix B has the same tables for the warm start.

A.1 Every joint angle

Table 3: MAE, cMAE, RMSE and r per coordinate on the envelope and on all inputs, the matched 15 first.

Gap: Every coordinate MOTOR outputs (the 31 of Appendix B's table): MAE, cMAE, RMSE and mean r on the matched envelope and over all scored inputs, the dagger for a reference range under 15°, the pooled matched 15, and the held-out subjects' rows apart; the warm start beside each.

Table 4: Per coordinate and activity.

Gap: MAE and cMAE of every coordinate per activity (walking, walking with trunk sway, squats, sit-to-stand, drop jump), with cycle-aligned waveforms of the hip, knee and ankle per activity.

A.2 Per subject group, activity and camera

Table 5: Per group: inputs, matched MAE and cMAE, vertical GRF MAE, median vertical r per foot, joint moment MAE, mean EMG r.

Gap: Rows: held out 7–11, training 2–5, each of the five activities and each of the five cameras (Cam0 to Cam4); MOTOR's values with the warm start's in brackets.

A.3 Ground reaction force

Table 6: Ground reaction force per activity on all three axes.

Gap: MAE in % body weight, vertical, anterior and medio-lateral, both feet summed as OpenCap scores them, per activity and held out apart; the share of the plates' stance frames each foot carries; the centre-of-pressure error; OpenCap's two-camera row beside it.

A.4 Joint moments

Table 7: Joint moments per coordinate and activity.

Gap: MAE in % weight × height for each of OpenCap's fifteen joint moments per activity, the knee adduction moment's early- and late-stance peaks in walking, against OpenSim's inverse dynamics on the lab's model; OpenCap's published values beside them.

A.5 Residuals

Table 8: The pelvis residual per group.

Gap: Per group: the median force and moment, the 95th-percentile force, and the share of frames within both limits (2% body weight, 1% weight × centre of mass height).

A.6 Muscle activations against EMG

Table 9: Each of the 16 EMG channels against MOTOR's activation of that muscle.

Gap: On–off agreement, Pearson's r and the lag at the cross-correlation peak for each channel (biceps femoris long head, semitendinosus, gastrocnemius medialis, soleus, gluteus medius, vastus lateralis and medialis on both sides, rectus femoris and tibialis anterior on the left), per activity and held out apart; the vasti symmetry AUC per camera against the instructed and the EMG-labelled asymmetry.

A.7 Time per step and repeatability

Table 10: MOTOR's wall time per step on one GPU.

Gap: Median, range and per-frame seconds of each step (fit levels 2 and 3, the whole-trial solve's movement, dynamics and muscles) over all scored inputs, the GPU named; two runs compared bit for bit.

A.8 Failures

Table 11: The inputs without a score, by cause.

Gap: Every input of the 720 that did not run or was not scored, by cause.

B The warm start's full validation

These tables are the per-frame chain's (Section 2.7): the fit, footing, residual reduction and two-stage static optimisation, which is now the whole-trial solve's warm start. They are not MOTOR's results. In the chain's final run of 7 November 2026, all 720 single-camera inputs were run; 715 were scored, and Section B.7 gives the rest by cause. "This run" below is that run; "before" and "the previous full run" are the chain's earlier full run of the same 720 inputs (700 scored), before changes to its contact model, its floor and its joints' ligaments, scored again with the same code. Every number is from paper/evidence/sim-body/labval-final-2026-11-07.txt, except the published solve times of Table 19.

B.1 Every joint angle

Table 12: MAE and centred MAE (degrees) per coordinate, previous full run → this run, and this run's mean r (†: the reference moves less than 15°). Envelope: 45° camera, no drop jumps (72 → 89 trials). All: every scored input (700 → 715). The matched 15 first; pro_sup has no shared zero, so only its cMAE.

	Envelope			All		
Coordinate	MAE	cMAE	r	MAE	cMAE	r
ankle angle r	6.41 → 6.39	3.18 → 3.19	0.89	5.97 → 5.96	3.94 → 3.92	0.92
ankle angle l	4.45 → 4.61	3.22 → 3.32	0.89	5.52 → 5.51	3.75 → 3.75	0.92
knee angle r	6.30 → 6.06	3.16 → 3.04	0.99	6.02 → 5.99	3.18 → 3.17	0.99
knee angle l	6.10 → 5.94	3.33 → 3.24	0.98	6.11 → 6.09	3.51 → 3.50	0.98
hip flexion r	8.17 → 8.03	3.90 → 3.82	0.98	8.91 → 8.89	4.16 → 4.11	0.98
hip flexion l	7.46 → 7.37	3.56 → 3.53	0.98	8.96 → 8.93	4.22 → 4.17	0.98
hip adduction r	4.20 → 4.20	2.81 → 2.73	0.52†	3.69 → 3.72	2.58 → 2.59	0.52†
hip adduction l	4.78 → 4.72	2.83 → 2.76	0.60†	3.48 → 3.52	2.50 → 2.52	0.57†
hip rotation r	4.10 → 4.07	2.62 → 2.62	0.43†	4.38 → 4.36	2.77 → 2.77	0.51
hip rotation l	4.53 → 4.41	3.19 → 3.06	0.43†	4.22 → 4.20	3.03 → 3.00	0.54
pelvis tilt	5.93 → 6.06	3.60 → 3.64	0.51	7.05 → 7.02	4.43 → 4.39	0.40
pelvis list	3.26 → 3.15	2.48 → 2.46	0.17†	3.10 → 3.11	2.22 → 2.24	0.16†
lumbar extension	4.66 → 4.56	3.15 → 3.11	0.64	5.36 → 5.32	3.37 → 3.35	0.75
lumbar bending	4.37 → 4.19	3.31 → 3.21	0.44	3.68 → 3.69	2.51 → 2.54	0.45†
lumbar rotation	3.67 → 3.67	2.14 → 2.12	0.58†	4.01 → 3.97	2.74 → 2.71	0.44
arm add l	3.79 → 3.98	2.33 → 2.34	0.70	6.62 → 6.58	5.18 → 5.12	0.70
arm add r	4.30 → 4.23	2.25 → 2.23	0.51†	6.22 → 6.16	4.45 → 4.40	0.64
arm flex l	7.92 → 8.03	3.11 → 3.19	0.93	12.59 → 12.46	8.72 → 8.59	0.90
arm flex r	6.77 → 7.00	3.09 → 3.19	0.93	11.32 → 11.20	8.65 → 8.52	0.90
arm rot l	8.19 → 8.01	4.41 → 4.50	0.64	12.48 → 12.35	9.91 → 9.78	0.67
arm rot r	7.25 → 7.03	4.27 → 4.09	0.67	11.31 → 11.18	8.74 → 8.61	0.68
elbow flex l	23.80 → 23.68	4.74 → 4.61	0.74	24.77 → 24.70	5.73 → 5.67	0.79
elbow flex r	20.96 → 21.12	3.07 → 3.00	0.80	22.79 → 22.74	5.84 → 5.76	0.76
mtp angle l	15.23 → 15.35	2.49 → 2.47	0.15†	15.73 → 15.74	2.19 → 2.19	0.10†
mtp angle r	15.17 → 15.22	2.58 → 2.55	-0.02†	14.99 → 14.99	2.05 → 2.06	0.00†
pelvis rotation	2.76 → 2.70	2.63 → 2.57	0.50†	2.66 → 2.63	2.45 → 2.42	0.48†
pro sup l	-	2.82 → 2.82	0.16†	-	3.63 → 3.61	0.25
pro sup r	-	2.44 → 2.55	0.18†	-	3.48 → 3.46	0.26†
subtalar angle l	9.18 → 9.24	7.02 → 6.88	-0.21	9.74 → 9.73	6.25 → 6.25	-0.07
subtalar angle r	9.02 → 9.16	6.26 → 6.26	0.04	11.44 → 11.39	6.21 → 6.21	0.04
matched 15, pooled	5.23 → 5.16	3.10 → 3.06		5.36 → 5.35	3.26 → 3.25	

The angles are the same chain's in both runs (GEM-X and the fit are unchanged); the small

differences come from the inputs each run has. The elbows are about 21 to 25° off with a centred error of 3 to 6° (straighter than the lab's in each of the 89 envelope trials; GEM-X's arm estimate, which the fit follows), and the toes (mtp) 15° off with a centred 2 to 3°: constant offsets, not tracking errors.

B.2 Per subject group, activity and camera

Table 13: This run per group (the previous full run's value in brackets): inputs, matched MAE and cMAE (degrees), vertical GRF MAE (% body weight, both feet), median vertical r per foot, joint moment MAE over OpenCap's 15 (% weight × height), mean EMG r over the channels. Cameras: LabValidation's numbering; Cam1 is the envelope's 45° camera.

Group	n	MAE	cMAE	GRF vert.	r R / L	Moments	EMG r
held out 7–11	396	5.13 (5.14)	3.02 (3.02)	16.28 (54.33)	0.77 / 0.79	2.41 (2.43)	0.24
training 2–5	319	5.62 (5.65)	3.54 (3.56)	16.15 (53.07)	0.78 / 0.77	2.42 (2.48)	0.22
walking	135	4.38 (4.40)	2.92 (2.93)	19.55 (76.11)	0.82 / 0.80	1.88 (1.79)	0.27
walking (trunk sway)	135	4.67 (4.67)	3.05 (3.05)	16.31 (84.14)	0.82 / 0.84	1.72 (1.53)	0.27
squats	87	5.79 (5.79)	3.26 (3.28)	6.32 (47.18)	0.53 / 0.47	1.58 (1.54)	0.17
sit-to-stand	90	5.39 (5.34)	2.92 (2.91)	11.58 (47.71)	0.68 / 0.80	1.55 (1.74)	0.20
drop jump	268	6.03 (6.03)	3.63 (3.63)	19.27 (32.21)	0.78 / 0.78	3.59 (3.73)	0.22
Cam0	144	5.21 (5.20)	3.14 (3.14)	14.95 (54.10)	0.75 / 0.83	2.44 (2.44)	0.23
Cam1 (45°)	141	5.62 (5.72)	3.33 (3.40)	15.50 (53.27)	0.73 / 0.81	2.54 (2.65)	0.22
Cam2	144	5.36 (5.36)	3.53 (3.53)	15.54 (52.72)	0.81 / 0.82	2.21 (2.31)	0.25
Cam3	143	5.35 (5.35)	3.20 (3.21)	15.96 (53.03)	0.81 / 0.70	2.44 (2.44)	0.23
Cam4	143	5.23 (5.23)	3.05 (3.05)	19.16 (55.68)	0.78 / 0.75	2.43 (2.45)	0.23

B.3 Ground reaction force

Table 14: Ground reaction force per activity, this run (previous run in brackets): MAE in % body weight on three axes with both feet summed, as OpenCap scored it; the share of the plate's stance frames the run's foot carries.

Activity	vertical	anterior	medio-lateral	3-axis mean	stance carried R / L
all	16.22 (53.77)	7.41 (6.45)	2.99 (3.19)	8.9 (21.1)	0.82 / 0.79
walking	19.55 (76.11)	12.03 (9.65)	3.02 (3.35)	11.5 (29.7)	0.82 / 0.75
walking (trunk sway)	16.31 (84.14)	10.86 (9.11)	3.49 (4.21)	10.2 (32.5)	0.83 / 0.85
squats	6.32 (47.18)	3.53 (2.56)	3.30 (4.17)	4.4 (18.0)	0.99 / 0.95
sit-to-stand	11.58 (47.71)	4.42 (4.74)	3.25 (3.51)	6.4 (18.7)	0.82 / 0.82
drop jump	19.27 (32.21)	5.61 (5.37)	2.54 (2.20)	9.1 (13.3)	0.76 / 0.71
OpenCap, 2 cameras	11.4	3.5	3.8	6.2	–

B.4 Residuals

Table 15: The pelvis residual this run leaves, per group: the median over trials of each trial's median force (% body weight) and moment (% weight × centre-of-mass height), the median of each trial's 95th-percentile force, and the mean share of frames within both limits (2% body weight, 1% weight × centre of mass height). No trial is within them on every frame.

Group	trials	force	moment	force, 95th pct.	frames within
all	715	7.0	8.90	24.4	21%
held out 7–11	396	7.1	9.31	23.9	21%
walking	135	7.8	10.99	23.0	6%
walking (trunk sway)	135	7.0	9.59	20.3	7%
squats	87	0.7	1.06	8.4	49%
sit-to-stand	90	0.2	2.82	8.8	46%
drop jump	268	10.9	9.32	85.5	17%

B.5 Muscle activations against EMG

Table 16: Each EMG channel against the chain's activation of that muscle, all 715 inputs: the share of frames on which both are on or both off, Pearson's r, and the lag (s) at the cross-correlation peak.

Muscle	on/off	r	lag	Muscle	on/off	r	lag
bflh l	0.65	0.13	-0.130	semiten r	0.70	0.23	-0.034
bflh r	0.65	0.13	-0.144	soleus l	0.70	0.26	-0.030
gasmed l	0.70	0.23	-0.009	soleus r	0.69	0.28	-0.041
gasmed r	0.69	0.26	-0.022	tibant l	0.59	0.22	+0.031
glmed1 l	0.51	0.13	+0.051	vaslat l	0.72	0.40	-0.003
glmed1 r	0.52	0.15	+0.040	vaslat r	0.72	0.42	-0.018
recfem l	0.56	-0.08	-0.002	vasmed l	0.72	0.36	-0.024
semiten l	0.66	0.18	-0.053	vasmed r	0.76	0.41	-0.033

Table 17: OpenCap's vasti symmetry index in squats: the AUC of the chain's index against the asymmetry the instructions asked for and against the EMG's own label (index over 1.15), previous run → this run. OpenCap: 0.83 against the EMG label.

Camera	AUC, instructed	AUC, EMG label
Cam0	0.47 → 0.84	0.64 → 0.75
Cam1	0.52 → 0.72	0.08 → 0.62
Cam2	0.60 → 0.95	0.52 → 0.84
Cam3	0.75 → 0.90	0.65 → 0.83
Cam4	0.53 → 0.81	0.25 → 0.48
all	0.56 → 0.73	0.45 → 0.66

B.6 Time per step

Table 18: The chain's wall seconds per step on one NVIDIA L4, over the 715 runs: median, range, and the median per frame (clips of 30 Hz).

Step	median s	min s	max s	s per frame
fit	21.825	14.483	623.317	0.303
set-up: physics model, GPU arrays, filtered path	4.697	3.878	32.851	0.061
footing	4.669	1.758	35.312	0.062
residual reduction, iterations	19.862	5.910	392.306	0.246
residual reduction, tracking and per-frame caps	4.603	1.714	45.888	0.061
checks: residual record, feet on the floor, tracking	2.120	1.111	17.879	0.028
static optimisation, stage 1 (ground reaction)	3.003	0.114	28.318	0.034
static optimisation, stage 2 (joint limits and muscles)	2.296	1.169	22.356	0.032
inverse dynamics, contacts and muscle paths	1.120	0.356	12.459	0.013
writing records and force measures	0.264	0.042	2.639	0.003
saving arrays	0.005	0.003	0.012	0.000
total	66.214	34.843	788.668	0.871

Table 19: Wall time per trial, the chain against published solves. OpenCap's solves are from its LabValidation release (80 muscles, windows of 0.99 to 1.68 s, solve only, hardware not recorded).

Chain	Clip	Wall time
OpenCap solve [6]	walk, drop jump, squat	4196, 4096, 2635 s
Moco, MocoTrack [4]	gait	about 130 min
OpenSim CMC [3]	gait	about 10 min
Warm start (chain), one L4	LabValidation, 715 trials	66 s median (35–789 s)

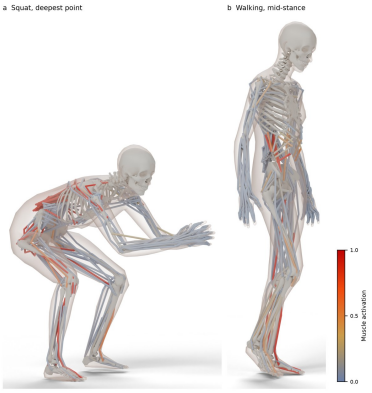


Figure 3: The warm start's MyoFullBody inside the person's SOMA skin, drawn from the chain's own saved LabValidation results: (a) a squat at its deepest point, (b) walking at mid-stance; each muscle coloured by its activation. The chain's output, not MOTOR's.

B.7 Failures

Table 20: The 5 of 720 inputs without a score in the chain's run, by cause.

Cause	Inputs
The GPU fit did not stop in 3000 iterations (subject 10, squats1, two cameras)	2
A frame's fit did not converge in 300 steps (two drop jumps)	2
Passed its job's time limit (subject 10, squats1, a third camera)	1
Total	5

C Joint definitions

The matched comparison scores fifteen joint angles, the coordinates that have a counterpart in OpenCap Monocular's set: ankle angle, knee angle, hip flexion, hip adduction and hip rotation on both sides, and pelvis tilt, pelvis list, lumbar extension, lumbar bending and lumbar rotation. The full technical validation scores also the shoulders (arm flexion, adduction and rotation), the elbows, the forearms' pronation–supination, the subtalar and metatarsophalangeal angles and pelvis rotation (Appendix B). Each angle is read from MyoFullBody's own coordinates after the fit; nothing is calibrated or regressed against the reference. The hip, knee, ankle, subtalar, toe and elbow angles are MyoFullBody's own coordinates of the same name. The pelvis and lumbar angles are OpenSim's intrinsic Z, X, Y angles of the model's own body frames, as the lab's model defines them, and the shoulders are the same angles of each humerus on the torso. Pronation–supination has its zero on another forearm pose in each model, so it is scored only by cMAE and r. The scoring follows the author's earlier work [38]: one sub-frame time shift of the reference per trial, found on the centred residual of both sides' hip, knee and ankle flexion; MAE, cMAE, RMSE and r; a dagger where the reference moves less than 15°; and the per-subject bootstrap of 20,000 draws.

D Model and solver parameters

Table 21: The bone groups: each SOMA segment, the MyoFullBody bone that carries its rotation, and the bones the group covers. Every sided group exists on both sides; 7 middle and 7 sided groups make 21, of which 20 drive their frame bone.

SOMA segment	Frame bone	Bones in the group
Hips	pelvis	pelvis, sacrum
Spine1	lumbar 4	lumbar 5 and 4
Spine2	lumbar 1	lumbar 3 to 1
Chest	torso	torso, thorax, cervical spine
Neck1	neck	neck
Neck2	neck (drives nothing)	–
Head	head	head
Shoulder	scapula	clavicle, scapula
Arm	humerus	humerus
ForeArm	ulna	ulna, radius
Hand	lunate	carpals, metacarpals, phalanges
Leg	femur	femur
Shin	tibia	tibia
Foot	calcaneus	talus, calcaneus
ToeBase	toes	toes

Table 22: MOTOR's parameters in one place.

Part	Parameter	Value
Model	muscles, bones, joints	416, 86, 132
	joints: root, coupled, free	1, 36, 95
Landmarks	defined, used by the fit	102, 84
Fit	markers' goal λ_m	2.1169
	skin goal λ_s	10^3
	most steps per frame	300
Level 1	segment end's reach	40 mm
	held-out rule	check bodies within 10% of tuning bodies
Level 2	sampled frames	about 30
	landmark's and inside joint's reach	3 cm
Whole trial	spline	uniform cubic B-spline, one control point per frame
	smoothing cut-off	6 Hz
	contact band	2 cm over the floor
	foot stiffness	6×10^6 N/m
	dissipation, friction, slip transition	2.0 s/m, 0.8, 0.05 m/s
	effort share	0.1 of the fit's cost at the warm start
	residual limits	2% body weight; 1% weight × centre of mass height

E Disclosure

The code of MOTOR and of its validation was written by a coding agent (Claude Code) under the author's direction and review; this draft was written the same way. No result was hand-picked. The only settings chosen on LabValidation (the chain's three contact settings and box clearance, and the whole-trial solve's contact settings and effort share) were chosen on subjects 2 to 5 against their force plates, never against joint angles; subjects 7 to 11 are held out and reported apart.